Timing repair with the Resizer

What it does Changes cells and buffering to reduce setup and hold violations, using OpenSTA to measure the effect of each round of repair.

Remember this Measure timing, find a bad path, change the circuit, and measure again. WNS and TNS are the scores that guide this loop.

How WNS and TNS guide repair

WNS identifies the worst negative slack. **TNS** adds negative endpoint slacks. The command `repair_timing` uses repair heuristics to improve these metrics. A heuristic is a practical search strategy, not a guarantee of the global optimum.

Setup repair means making late data faster

The Resizer targets the worst setup path and then works on other violating endpoints to reduce total negative slack. Depending on the library and settings, it can use these changes:

Gate sizing Replace a cell with a stronger equivalent to drive its load faster. A larger cell also adds area and input capacitance.

Buffering or removing buffers Split a large load or long wire into better-driven sections; remove an unnecessary buffer if its extra delay hurts.

Cloning or load splitting Give a critical load a less heavily loaded driver by duplicating suitable logic or reorganizing buffering.

Pin or threshold-voltage swaps Use a faster equivalent input pin or an available faster cell variant. Only legal, function-preserving changes are allowed.

Hold repair means slowing early data

If data arrives too early, hold repair typically inserts delay buffers in the **data path**. The extra minimum delay can fix hold, but consumes setup margin. OpenROAD normally repairs setup first and guards setup while inserting hold buffers.

The repair loop

Repeat until timing targets are reached, progress stops, or area, buffer and iteration limits prevent further repair. The tool may undo changes that fail its improvement criteria.

Inputs A timed physical design, libraries, constraints and parasitics.

Outputs A modified netlist and cell placement needing legalization, routing updates and timing verification.

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Why improving only WNS is not enough

Suppose three endpoint setup slacks are:

$$\text{Before:} \quad -0.20,\ -0.10,\ -0.05\ \text{ns}$$

WNS is -0.20 ns and TNS is -0.35 ns.

Repair the first endpoint so its slack becomes $+0.02$ ns:

$$\text{After:} \quad +0.02,\ -0.10,\ -0.05\ \text{ns}$$

WNS improves to -0.10 ns and TNS to -0.15 ns. **Two endpoints still fail.** Further repair is required. In a real circuit, one change can affect several endpoints at once.

A setup fix and a hold fix

Setup example A path has a 1.00 ns allowed arrival and arrives at 1.12 ns: slack $= -0.12$ ns. A suitable stronger driver reduces arrival to 0.96 ns: slack becomes $+0.04$ ns.

Hold example The earliest allowed arrival is 0.12 ns, but data arrives at 0.08 ns: slack $= -0.04$ ns. A delay buffer adds 0.06 ns: arrival becomes 0.14 ns and hold slack becomes $+0.02$ ns. Setup must be checked again.

Both are simplified teaching examples; real cell and wire delays depend on load, slew and operating conditions.

Commands to recognize

With CTS complete, propagated clocks and valid parasitics, the essential repair commands are:

```
repair_timing -setup -repair_tns 100
```

```
repair_timing -hold
```

`-repair_tns 100` targets all violating endpoints; `0` targets only the worst endpoint. The number is a percentage of endpoints to attempt, not a promised percentage improvement. Check `help repair_timing` in your installed version.

When can you claim timing closure

Re-legalize changed cells, update routes and parasitics, then rerun setup and hold analysis. Check all required timing corners, constraints and electrical limits. A clean pre-route estimate can fail after real wire delays are included. Area, power or congestion may also limit repair.

What to say to your professors

"OpenSTA measures WNS and TNS. The Resizer improves them by trying gate and buffer changes on critical paths. Setup repair speeds up late data; hold repair delays early data. We repeat analysis and repair with updated physical information until all required checks pass."

Sources and code OpenROAD Gate Resizer documentation, Repair Timing. Code: `src/rsz/src/Optimizer.cc`, the setup policies in `src/rsz/src/policy/`, and `RepairHold.cc`. Command behavior checked against local source.